



Figure 9.6-3 Soft keyboard.

|                                                                                   |                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Note !</b>                                                                                                                                                                      |
|                                                                                   | <p><b>Alternatively, a standard USB keyboard and mouse, can be used to interface with the HMI. These connect to a USB extender port in the rear of the operator panel E0_5</b></p> |



Figure 9.6-4 The external USB port on the HMI panel.

## HMI Operational Levels

The HMI software has been compiled from industry standard software packages. To ensure that the possibility of corruption to the operation program is minimised, access to certain software programs is limited by the use of passwords.

### Log In Screen

The computer has been set up to allow monitoring of usage by different operators and for production purposes, shift patterns.

On start-up and following the 'initial boot screen', a SYSTEM screen will be shown and the user must touch the flashing LOGIN button to allow logging on.

At this time of this login sequence, the user will be asked to enter their login name and password.

All other normally used screens are accessible from this screen. There are some screens that are rarely used and these are only available via the specific screens that they are attached to.

Use the keyboard to enter the user's login name and password.

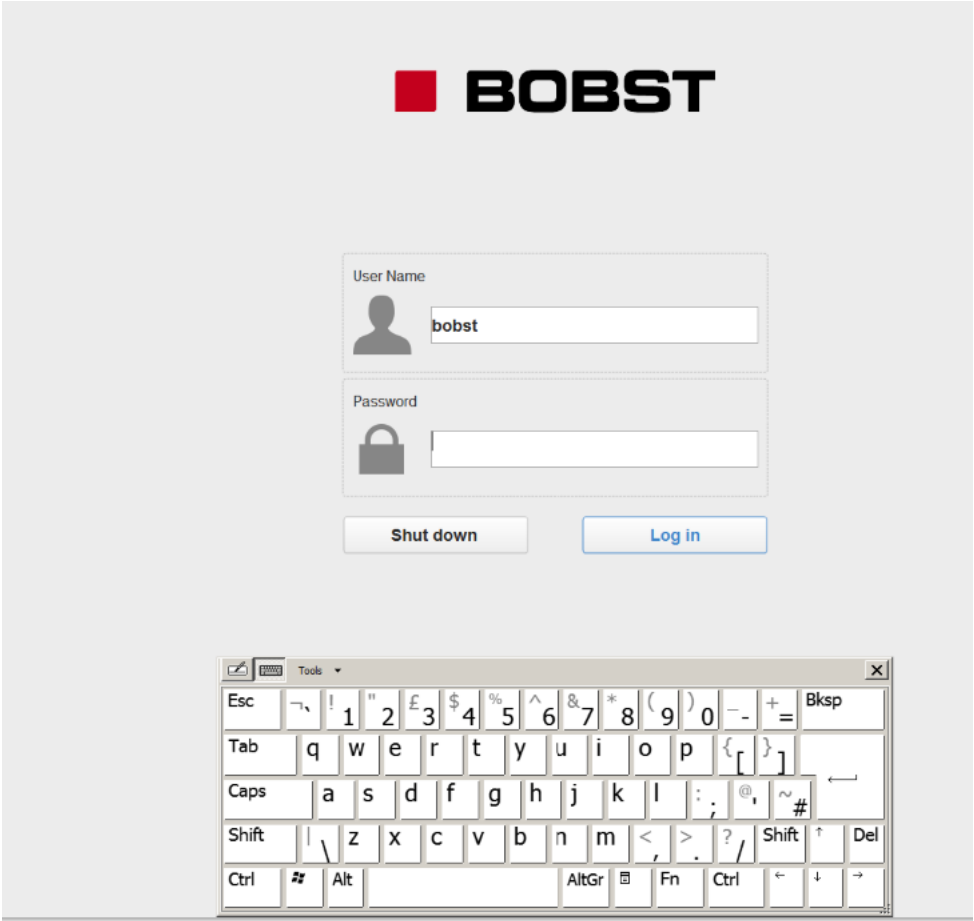


Figure 9.6-5 User Log in screen.

Standard Passwords

The standard ‘factory set’ passwords for operators are:

| Login name: | Login password: |
|-------------|-----------------|
| Operator 1  | Operator 1      |
| Operator 2  | Operator 2      |
| Operator 3  | Operator 3      |

# 9.7 Process Chiller Unit

The chiller units are located on the winding cart for the purpose of supplying a coolant to the machine's cooled process rollers.

The coolant, a glycol and water mixture are used to chill the process drum and roller to a temperature set between a range in degrees C. This range is as per the machine specification.

The chiller and associated valving operation is controlled from the individual chiller control system and is interfaced to the main machine plc.

It is interlocked to change from the “Standby” mode into the “Cool” mode when the vacuum pumping system is in the “vacuum mode”.

The mode is changed to “Heat” to allow the process rollers to be heated back to ambient before venting the machine. If the process drum and roller are not heated up, condensation will occur on them.

The chiller is supplied with customer tower/process water to allow their operation. Pressure and temperature sensors are used to monitor this water supply.

Monitoring of the chiller performance is provided from the CHILLER DIAGNOSTIC screen.

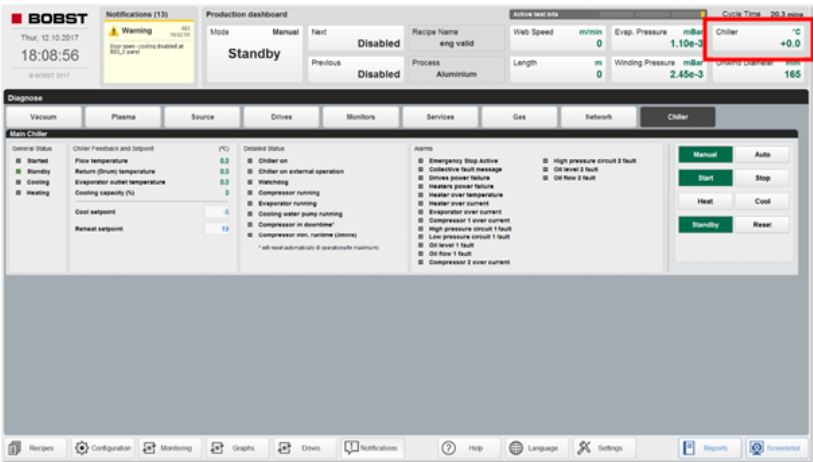


Figure 9.7-1 CHILLER DIAGNOSTIC HMI screen.

The chiller process temperature is set by the user either from the SETUP RECIPE interface screen or from the CHILLER DIAGNOSTIC screen.

Touching the highlighted set point will allow the user to change the set-point within the operational range of the chiller.

Caution !

Overriding of any sensors will invalidate any warranty agreement

Reference should be made to the vendor manual for explanations of the interlocks/faults indicated.

The system uses a mixture of ethylene glycol and water as its coolant.

Testing the Coolant

The coolant, when measured with a hydrometer must give a specific gravity of 1.080 at 68°F (20°C), which gives a freezing point of -30°C. This is the normal mixture for chillers rated to –15 lowest temperature.

A specific gravity less than 1.080 may cause the coolant to freeze and damage the chiller.

A specific gravity higher than 1.080 will delay the cooling process to taking place and in fact the process may not reach its operating temperature.

The coolant mixture should be tested as per the recommendation in the preventive maintenance schedule. It should be tested in the following manner.

1. Fill a container with a sample of the coolant taken from the cooling system. This can be drained from one of the two drains found under the chiller, near to the process in/out valves. The container should be inches (254 mm) deep and at least 3 inches (75mm) wide.
2. Place a Hydrometer in the sample and allow it to settle. The specific gravity level can be read off the scale. Where the liquid bisects the scale is the value.
3. If the value is greater (i.e. 1.130) add water to the top up gravity fed tank. Recirculate the coolant and retest a sample.  
In fact it is very rare to find a mixture change from its original specific gravity value.
4. If the number is lower (i.e. 1.040) add Ethylene Glycol to the mixture.

|                                                                                   |                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>Caution !</b></p> <p><b>The chiller heat exchange will be damaged if the mixture freezes due to incorrect coolant mixtures.</b></p> |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

If the cool down time of the chiller is excessive it could be due to the mixture going 'mushy' prior to freezing.

### Draining the Process Lines from the Chiller to the Rollers

The cooling medium can be drained from the cooling system by opening valves located under the cart floor.

The amount of cooling medium in the system will vary from machine to machine. It is usually in excess of 50 litres and therefore containers are required when draining the complete system.

It is suggested that the medium should also be removed from here when a sample is required.

### Glycol Specification

The chiller unit uses a medium made of a mixture of ethylene glycol and water. The quality of the ethylene glycol is specified in the consumable section of this manual. [See chapter Section 10.0 Process Consumables on page 504 for details.](#)

### Filling the Chiller and Bleeding

It is necessary to prime the water / glycol system in order to purge all of the air.

Recommended Priming Procedure is explained in the vendor manual supplied.

### Losses of Coolant from the System

There will be a natural evaporation of the coolant over a long time period. Checks should be made weekly on the level in the tanks. Addition of liquid must be made as required.

If the coolant volume changes for any reason and the upper level switch in the storage tank is not energised, the main circulation pump will not operate.

## 9.8 Machine Air System

Air is supplied to various parts of the machine from the central air regulator located on the vacuum booster pump support steel frame.

There is an isolation valve and regulator located at this point.

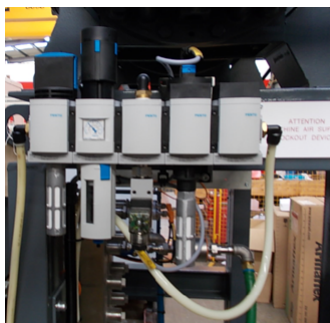


Figure 9.8-1 The air isolation and regulation unit.

The specification for the pressure and volume requirements can be found on the pneumatic schematic drawing provided.

Air is used for the following functions;-

- Operation of the vacuum isolation valves (Fail-safe to Close on loss of power)
- Operation of source viewing flap
- Operation of water isolation valves
- Operation of DTR1 glycol supply line valve (Option)

The air supply is monitored at all times by a pressure switch and any faults are recorded by the computer and displayed upon the alarm screen.

|                                                                                     |                                                                                                            |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
|  | <b>Caution !</b>                                                                                           |
|                                                                                     | <b>Ensure that the air supply is clean and dry. Dirty air will cause problems in the pneumatic systems</b> |

## 9.9 Machine Water System

Water is provided to the machine as a cooling medium. It is used to cool the following:-

- Source busbar system
- Diffusion pumps
- Vacuum pumps
- Evaporator source enclosure
- Plasma Treater Electrodes
- Chiller Condenser
- Drum Shields and frame
- Evaporator contact post & blocks

The specification for the water temperature and flow requirements can be found upon the water schematic drawing supplied in the documentation CD/DVD.

Isolation valves are provided at the incoming supply and return lines, to allow for complete isolation.



Figure 9.9-1 Water isolation lines and sensors.

Pressure sensors are provided on all the water manifolds to monitor the water supply to the machine. These sensors are monitored by the computer for low pressure and loss of pressure differential across the feed and return lines. In either situation, an alarm will be indicated.

In addition, flow sensors are provided on some equipment to monitor the water supply to the equipment. These sensors are monitored by the machines control systems for low flow. In situations of low flow, an alarm will be indicated.

The manifold is located in the area shown on the machine plan below.

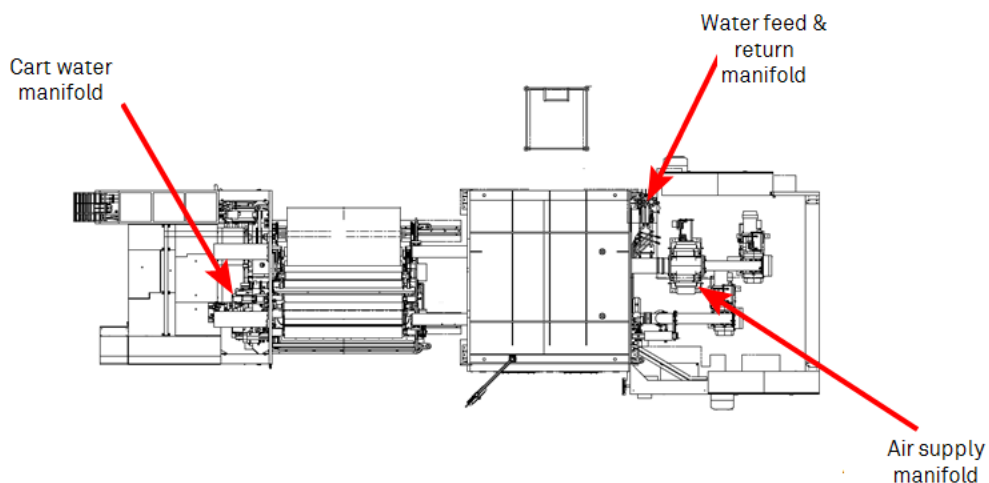


Figure 9.9-2 Plan view of machine indicating water manifold location.

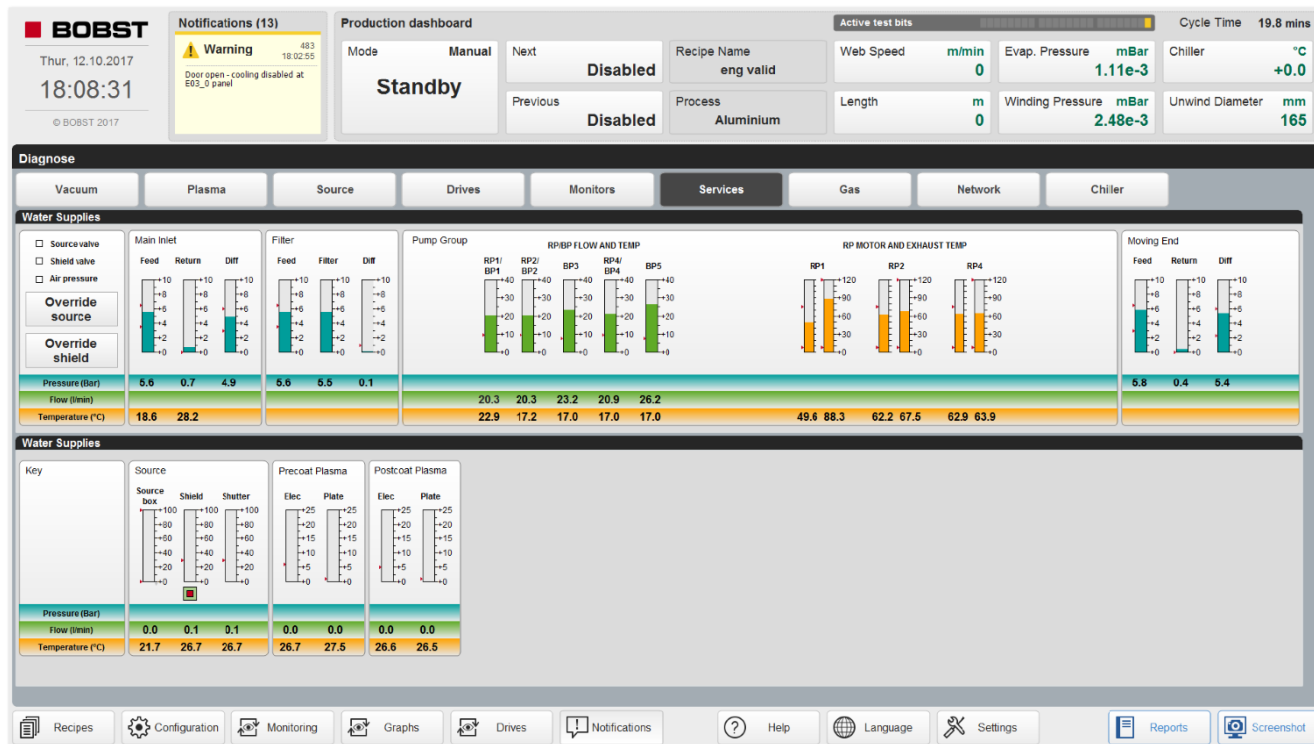


Figure 9.9-3 SERVICES DIAGNOSTIC HMI screen.

This screen is provided to allow the user to check on the status of the water system on the machine.

|  |                                                                                                                                                       |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>Warning !</b></p> <p>Do not modify any circuits on the machine without written consent of Bobst Manchester Ltd.</p> |
|  | <p style="text-align: center;"><b>Caution !</b></p> <p>Overriding of these sensors will invalidate any warranty agreement</p>                         |

## Water Isolation to Internal Cooled Circuits

To assist in the reduction of condensation on the machine, the water to the coating shield, plasma system and evaporation source is isolated once the machine has been opened fully.

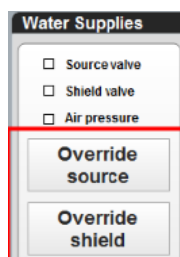


Figure 9.9-4 Water HMI diagnostic screen highlighting isolation valves to internal components..

|                                                                                   |                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Note !</b>                                                                                                                                                                                                                                         |
|                                                                                   | <p>At the maintenance log level, these valves can be activated by touching the icon on the screen, when the winding cart is fully retracted.</p> <p>If the chamber is closed and the rotary pump RP1 is not running, no water will be circulated.</p> |

A strainer is fitted in the supply line to the machine to ensure that large debris does not enter cooling pipes. This should be cleaned on a regular basis and should not be left out of the circuit at any time.

|                                                                                   |                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Warning !</b>                                                                                                                                                                                                                      |
|                                                                                   | <p>Do not remove the element from the strainer as this will invalidate any warranty agreement</p> <p>Do not install valves in the return lines of the water circuit as this could lead to accidental damage to water cooled items</p> |

|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Note !</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                                    | <p>If the temperature falls below 22 C or above 28 C, a warning will be indicating to the user that there is a temperature issue with the supply.</p> <p>At 30C, the machine will not allow the evaporators to be switched on, and will stop the process if running.</p> <p>If the temperature continues to be at this level, for a further 10 minutes, the pumping system will shut down.</p> <p>The system will not rest until the supply water temperature is fallen below 25C</p> <p>Minimum flow rates are pre-set to the values given on the water schematic drawing</p> |



## 9.10 Coating Shield System

The coating shield is located on the winding cart can be lowered for cleaning and adjusting the width of the drum edge shields, or to clean substrate out from behind it, or when threading substrate through the winding mechanism.

Control push buttons are located on the front of the winding cart, on the electrical panel to allow this operation to take place when the winding cart is fully retracted.

|                                                                                   |                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Note !</b>                                                                                                                                                                                                                       |
|                                                                                   | <p><b>The shield cannot be lowered or raised if the E-stop circuit has been activated or the Traverse Lockout is in the OFF position.</b></p> <p><b>The shield cannot be raised if the drum cleaner foot switch is engaged.</b></p> |

The following sequence should be followed to lower the shield;

1. Press the lower push-button until the shield is fully lowered. It will stop automatically.
2. Turn the LOCKOUT isolator through 90 degrees and place the users padlock on it.

The shield is then safe to work upon

|                                                                                   |                                                                                                                                                                         |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Warning !</b>                                                                                                                                                        |
|                                                                                   | <p><b>Do not become trapped between the coating shield and the machine at any time. Turn the isolator after lowering the shield and lock it out with a padlock.</b></p> |



Figure 9.10-1 Example of an operator lowering the coating shield.



Figure 9.10-2 Example of an operator placing a lock on the traverse isolator.

The following sequence should be used to raise the shield;

1. Remove the padlock off the LOCKOUT isolator and turn it to the on position
2. Ensure that the area is clear.
3. Press the raise push-button to raise the shield.

The drum coating shield is provided to allow adjustment for different widths of substrate to be coated without coating the drum with the aluminium

The shielding is manufactured from water-cooled and non-cooled sections.

Some highlights of the system are;-

- The concave section that locates close to the drum to stop the coating of the drum.
- The aluminium edge shield section has an extended plate that is a non-cooled.
- The non-cooled sections are located away from the evaporators that are heated up.



Figure 9.10-3 Example of a water-cooled frame in its lowered position.

The machine is specified for maximum and minimum coating widths. Sections of shielding are provided to allow covering of the drum for any substrate width between these minimum and maximum widths.

## Cleaning the Coating Shield

Before attempting to change the shields, the shield requires to be cleaned to allow movement of its components.



Figure 9.10-4 Example of an operator cleaning the coated shutter with a scraper tool.



Figure 9.10-5 Example of an operator cleaning the cooled frame with a scraper tool.

|                                                                                     |                                                                            |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|  | <b>Caution !</b>                                                           |
|                                                                                     | Use relevant personnel protection equipment when carrying out these tasks. |

|                                                                                     |                                                                                                 |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|  | <b>Note !</b>                                                                                   |
|                                                                                     | Attempting to move the shield without cleaning it first will cause problems with items binding. |

## Adjusting Drum Coating Width Shields

The following instructions must be used when changing the shields.

1. Lower the shield assembly as described previously.
2. Lock out the movement isolator.
3. Clean the shield area.
4. Isolate the water supply at the manifold on the winding cart.
5. Release the water supply couplings, located within the shield.



Figure 9.10-6 Example of an operator releasing the self-sealing water couplings.



**Note !**

**Please be aware that if debris is in the water line, the coupling may not self-seal. Therefore, it is advised to isolate the water at the manifold on the winding cart.**

6. Slide the shield in towards the centre of the coating shield



Figure 9.10-7 Example of an operator removing the non-cooled section.

7. Withdraw the water hoses back through the coating shield to allow them to be pulled clear.





Figure 9.10-8 Example of an operator passing the water hose through the side of the shield arm.

8. Remove the shield away from the machine.
9. Reinstall a replacement edge shield in reverse to the instructions above.
10. Reconnect the water and check the water diagnostic screen to ensure that water flowing is indicated.
11. Raise the assembly using the push-button.
12. With the substrate in the machine (cantered) and under tension, mark the inner edges of the substrate with a "marker pen" and inch the drive system. This will indicate the amount of substrate that will be covered by the shields and therefore uncoated.
13. If the clear edge is excessive or too small, adjust the shield slightly, by hand. The clear edge should be normally 6mm.



Figure 9.10-9 Example of a disconnected self-sealing coupling.

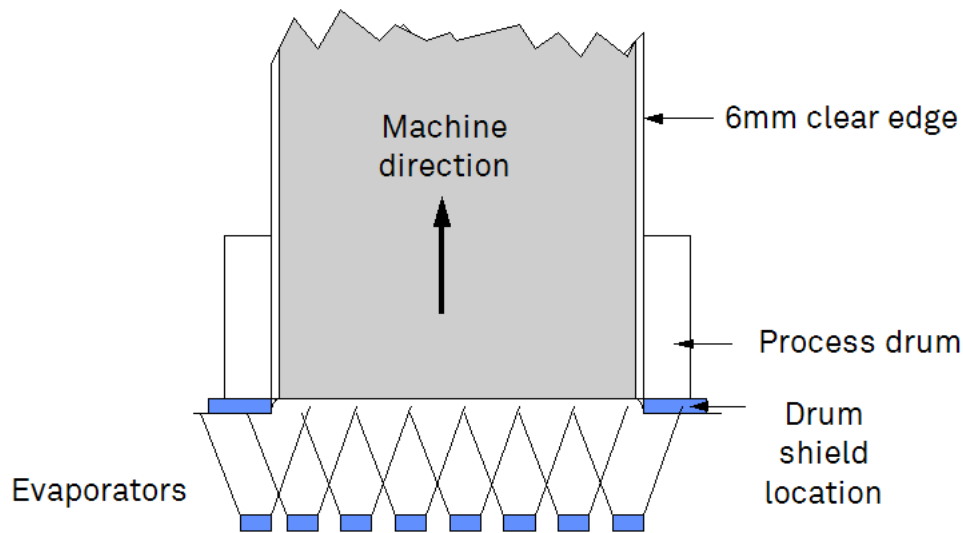


Figure 9.10-10 Position of drum side shields in relationship to substrate edges.

**Warning !**

**DO NOT BECOME TRAPPED BETWEEN THE COATING SHIELD AND THE MACHINE AT ANY TIME. TURN THE TRAVERSE ISOLATOR AFTER LOWERING THE SHIELD AND LOCK IT OUT.**

Testing Water Flow Following Shield Change

Following any change to the drum edge shield it is strongly suggested that the water flow is checked to make sure that there are no water leaks and that the flow is correct.

Log into the HMI system at the supervisor/maintenance level and access the water diagnostic screen.

Touching the screen valve icon, when the winding cart is fully retracted, will open the water isolation valve and allow flow to take place.

The small triangle in the drum shield line will be white if the flow is above 5 litres/min.



On the screen, the actual flow through the shutter and coating shield will be shown.

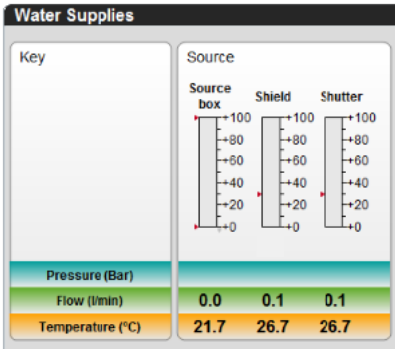


Figure 9.10-11 HMI water diagnostic screen.

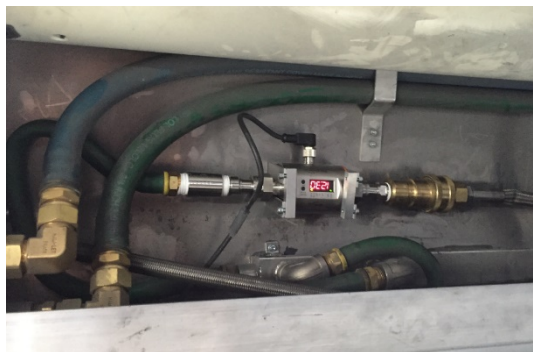


Figure 9.10-12 A shield edge flow monitor displaying flow rate.

## Adjustment/Removal of Coating Shield Process Sine Waves

To reduce heavy deposition areas on the substrate at high evaporation rates, a sine wave shield has been designed to be attached to the coating shield. This is split into a number of smaller sections for ease of handling.



Figure 9.10-13 A coating shield with a full set of sine waves fitted.

Dependent upon the deposition rate (wire feed rate); the sine waves selected will vary.

| Wire Feed Rate           | Upper Sine Wave Fitted (As viewed with the coating shield Lowered) | Lower Sine Waves Fitted (As viewed with the coating shield Lowered) |
|--------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------|
| <60 cm/min               | No                                                                 | No                                                                  |
| Between 60 and 90 cm/min | No                                                                 | Yes                                                                 |
| > 90 cm/min              | Yes                                                                | Yes                                                                 |

Slotted countersunk fixing screws have been provided to ensure that they can be cleaned of aluminium and removed.



Figure 9.10-14 A section of sine wave.

On the rear of the shield, at the mating surface, a piece of 0.5mm thick graphite tape is used to ensure good heat transfer from the copper shield to the cooled zoning shield. It is essential that this material is used at all times.

**Caution !**

Do not operate the machine without a layer of graphite between the mating surfaces or without all the required screws being installed and tightened.  
Failure to do this will result in thermal damage to the parts.



Figure 9.10-15 Example of an operator removing the shield.





Figure 9.10-16 Example of an operator showing the graphite tape on the rear of the shield.

## Light Deposition Coating – Blanking Plates

There may be times, such as when light optical density coatings are required for solar film, etc., when the sine waves should be removed and replaced with a straight edge Blanking Plate. These should also have a graphite mating surface.

|                                                                                    |                                                                                                                                                              |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>Caution !</b></p> <p>Always use either a sine or a copper blank plate to stop potential damage to the mating surfaces.</p> |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                     |                                                                                                                                |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>Note !</b></p> <p>Bobst Manchester Ltd. supply these parts when the machine is installed</p> |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|

## Coating Shield Edge Shield support Strip Removal

The water cooled drum edge shields are located in the machine by support strips.

Under normal operation, evaporated aluminium covers these support strips. There could be a requirement to remove the strips for cleaning or replacing purposes.

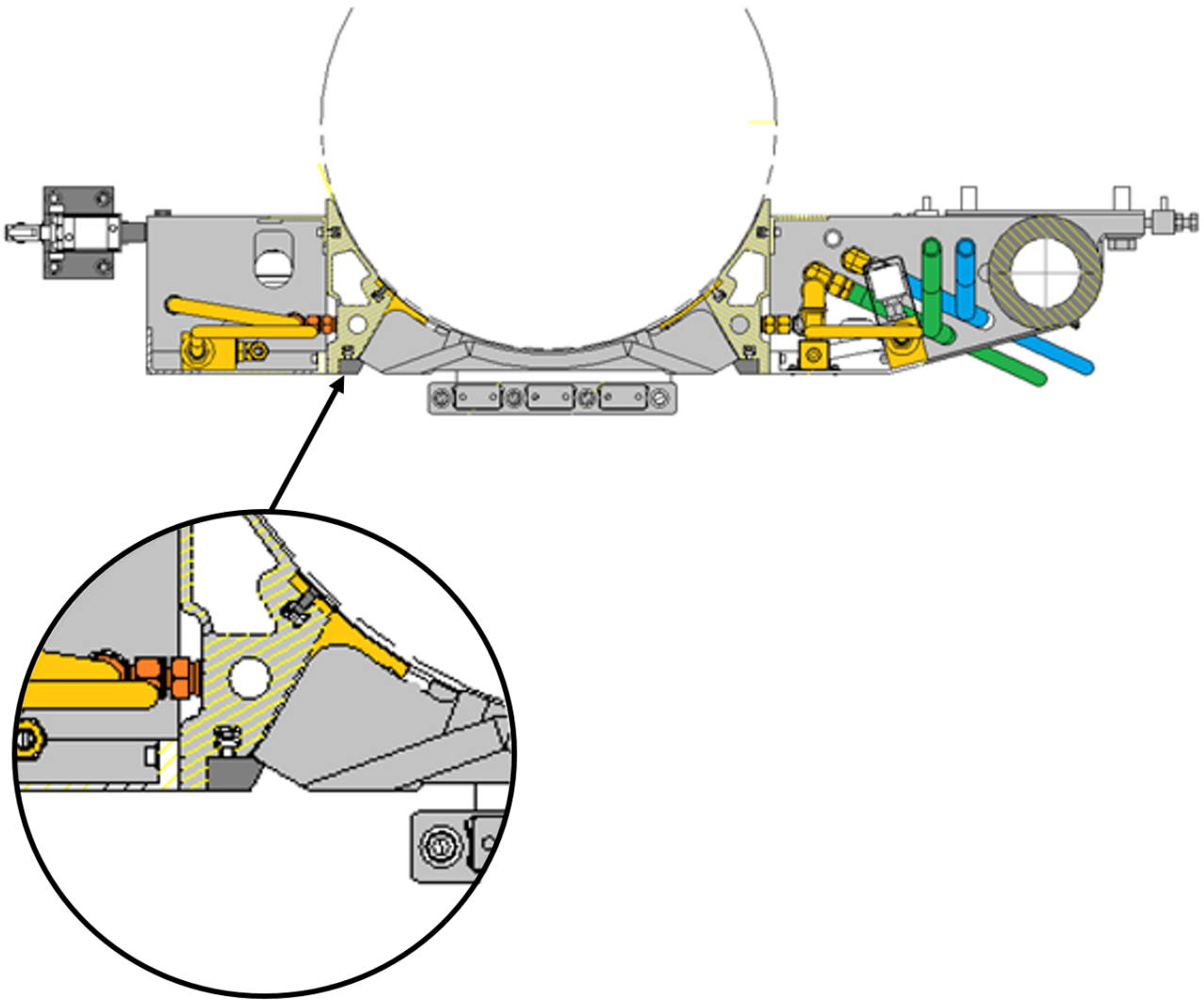


Figure 9.10-17 Cross section of process drum area, showing the location of the removable support strips (BML part # 417G287).

These strips can be removed to allow cleaning behind them, as follows.

1. Lower the coating shield to the vertical position
2. Loosen the hexagonal bolts using a socket. At this point do not remove them

